

Title of Research Project: Integrating Physical Activity and VR-Based Mindfulness to Support Stress and Cardiometabolic Health in Breast Cancer Patients: A Feasibility Study

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Project Summary: Breast cancer (BC) patients often experience significant psychological distress and cardiometabolic health risks during and after treatment, impacting their overall quality of life. Women with BC, particularly those receiving certain treatments (left-sided radiation therapy and endocrine therapy), may have a higher risk of developing hypertension, dyslipidemia, and diabetes. This study aims to evaluate the feasibility and effects of a novel 12-week intervention combining physical activity and virtual reality (VR)-based mindfulness training in patients undergoing chemotherapy for early-stage breast cancer.

This study will recruit 10 participants from the Stony Brook Cancer Center, assessing feasibility through recruitment, retention, adherence, and satisfaction. The intervention will include a patient-centered, home-based aerobic and resistance exercise program, alongside immersive VR mindfulness delivered three times weekly. Outcomes will include changes in perceived stress, cardiometabolic indicators (e.g., blood pressure, body composition, exercise endurance), and biomarkers (e.g., cortisol, C-reactive protein, and lipids).

Through an interdisciplinary and patient-centered approach, this project seeks to improve mental and physical health outcomes through exercise and immersive VR mindfulness. If successful, these results could inform scalable interventions that support survivorship and reduce associated comorbidities in breast cancer care.